

Original Article

Genetic architecture underlying the parallel evolution of leaf ecomorphs in *Viburnum*

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ABSTRACT

In a neotropical lineage of the plant clade *Viburnum* (Adoxaceae) several leaf ecomorphs evolved independently and repeatedly as the group radiated through cloud forests of North and South America. Here, we focus on one pair of co-occurring sister species within this radiation with strongly contrasting leaf morphotypes and document the presence of phenotypically diverse and genetically admixed hybrid individuals in multiple hybrid swarms. Hybrid phenotypes are generally intermediate in form, but sometimes show parental or entirely novel and transgressive combinations of leaf traits, suggesting that parental leaf ecomorphs can be functionally and genetically dissociated. We used admixture mapping within hybrid swarms to investigate the genetic architecture of key traits comprising these leaf ecomorphs and uncovered loci proximal to known genes implicated in leaf development, including some that may alter multiple leaf traits simultaneously, potentially facilitating the emergence of leaf syndromes. We conclude that the shared genetic architecture underlying some traits, such as leaf size and marginal teeth, could promote the repeated evolution of these traits in concert, while low levels of genetic linkage between other leaf traits supports the hypothesis that selection promoted the repeated assembly of particular combinations of leaf traits as *Viburnum* radiated throughout the neotropics.

Keywords: admixture; evolutionary genomics; genetic architecture; leaf ecomorphs; morphological evolution

INTRODUCTION

A major challenge for evolutionary biologists is to explain the repeated assembly of particular combinations of traits into readily recognizable ‘syndromes’, ranging from pollination, dispersal, photosynthetic, or defence syndromes in plants (Fenster *et al.* 2004, Agrawal and Fishbein 2006, Edwards 2019, Dellinger 2020, Sinnott-Armstrong *et al.* 2020), to ‘ecomorphs’ in stickleback fishes and *Anolis* lizards (Losos 1992, Rundle *et al.* 2000). It is generally assumed that the parallel evolution of traits underlying syndromes reflects particular selective forces acting on underlying genetic variation (Sinnott-Armstrong *et al.* 2020). However, genetic architecture and development could play a critical role in enabling (or inhibiting) the emergence of particular trait combinations (Malinsky *et al.* 2015, Erickson *et al.* 2016, Wessinger and Hileman 2016, Funk *et al.* 2021). How can

these various causal factors be teased apart, how do they interact, and to what degree do they determine particular outcomes?

Here we investigate the genetic architecture underlying leaf trait ecomorphs in *Viburnum* using an admixture mapping approach. Admixture mapping is a method of genome wide association that allows interrogation of genotype-phenotype relationships by taking advantage of the natural diversity of recombinant genotypes generated by hybridization (Patterson *et al.* 2004). This approach has proven to be a powerful tool for untangling the genetic architecture of traits in a wide variety of organisms, especially organisms that cannot be crossed in the laboratory for traditional mapping studies (Winkler *et al.* 2010, Breilsford *et al.* 2017, Fuller *et al.* 2020) including long-lived, large, and woody plants (Lindtke *et al.* 2013, Bresadola *et al.* 2019, Müller *et al.* 2020). Hybridization is common in many

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plant taxa and can unveil extensive hybrid trait variation in the wild (Rieseberg *et al.* 2007, Lindtke *et al.* 2013, Sung *et al.* 2018, Singhal *et al.* 2020).

Viburnum is a clade of approximately 165 species of shrubs and small trees found on five continents but with a centre of diversity in eastern Asia (Winkworth and Donoghue 2005, Clement *et al.* 2014, Spriggs *et al.* 2015, Landis *et al.* 2021). *Viburnum* species exhibit extensive variation in leaf size, shape, margin, and trichome characters (Donoghue 1982, Schmerler *et al.* 2012, Spriggs *et al.* 2018). Many *Viburnum* species also readily hybridize, and artificial crosses have been used extensively to generate cultivars with desirable horticultural properties (Egolf 1956, 1962). Naturally occurring hybridization is also reported in *Viburnum* where ranges of closely related species come into contact (McAtee 1956, Donoghue 1982, Hara 1983, Winkworth and Donoghue 2004, Spriggs *et al.* 2019).

We recently documented a case of ‘replicated radiation’ in the *Oreinotinus* clade of *Viburnum* (Donoghue *et al.* 2022). This lineage radiated rapidly from north to south through the cloud forest regions of Mexico, and Central and South America, starting in the Miocene, roughly 12 Mya (Spriggs *et al.* 2015,

Landis *et al.* 2021). As they progressively occupied 11 different areas of endemism, they repeatedly evolved *in situ* a similar suite of leaf morphologies (Donoghue *et al.* 2022). Within nine of the 11 regions, species have evolved two or more of the following leaf syndromes: (i) large, pubescent, and with toothy-margins (LPT), (ii) small, glabrous, and with entire-margins (SGE), and (iii) intermediate-sized, glabrous, and with entire-margins (IGE). In this study we focus on one pair of sister species native to Chiapas, Mexico: *Viburnum jucundum* Mort. with LPT leaves, and *Viburnum lautum* Mort. with SGE leaves (Donoghue *et al.* 2022) (Fig. 1). Both species are tetraploids with $2n = 36$ and with very similar genome sizes ($2n \sim 12$ Gbp) (Moeglein *et al.* 2020).

The contrasting leaf ecomorphs of *V. jucundum* (LPT) and *V. lautum* (SGE) may promote differences in ecophysiology and environmental distribution between species. For example, larger, rounder, and more pubescent leaves, or leaves with less dissected margins, have thicker boundary layers, which translate into an increased risk of damage in freezing or in hot and dry environments (Gates 1965, Parkhurst and Loucks 1972, Wright *et al.* 2017). Woody plant leaves bearing marginal teeth

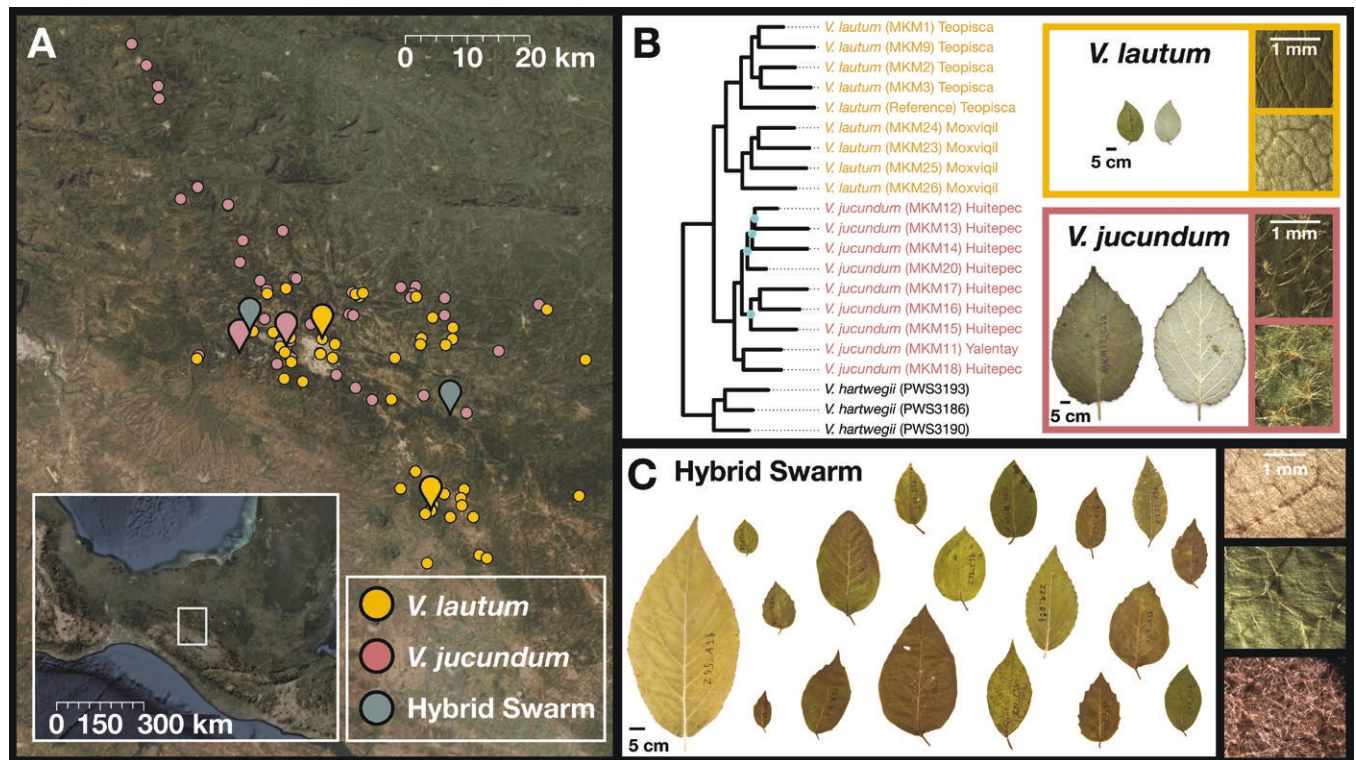


Figure 1. A, Collection localities of *V. lautum*, *V. jucundum*, and hybrid swarms across the central plateau of Chiapas, Mexico (region inset). Coloured circles indicate herbarium specimen collections while pins mark populations used for genetic and leaf morphological analyses. The northernmost yellow pin corresponds to the ‘Moxviquil’ population of *V. lautum* while the southernmost yellow pin corresponds to the ‘Teopisca’ population of *V. lautum*; the westernmost pink pin corresponds to the ‘Yalentay’ population of *V. jucundum* while the easternmost pink pin corresponds to the ‘Huitepec’ population of *V. jucundum*; the northwestern grey pin corresponds to the ‘Zinacantan’ hybrid swarm population while the southeastern grey pin corresponds to the ‘Yashtinin’ population of *V. lautum*. B, Phylogenetic tree (see Material and methods) depicting sister-species relationship between *V. lautum* and *V. jucundum* for samples included in this study with representative leaf images for both species plus images of adaxial (top images) and abaxial (bottom images) leaf surfaces showing variation in trichome density between species. All bootstrap support values are greater than 98 with the exception of nodes labelled with blue dots. Support for these labelled nodes does not ultimately change support for relationships between species. C, Selected leaves from various hybrid swarm individuals demonstrating variation in leaf size, shape, and margin characters with abaxial trichome density (images on right) varying from glabrous to densely pubescent, and trichome morphological variation from unbranched to stellate.

predominate in cooler environments and leaf teeth have been linked with multiple phenomena including leaf hydraulic supply and support, increased early-season photosynthetic rates, and increased leaf development within resting buds (Bailey and Sinnott 1916, Givnish 1987, Royer and Wilf 2006, Edwards *et al.* 2016). The relationship between trichome characters, physiology, and climate is variable. In addition to their role in modifying boundary layer dynamics, trichomes have been implicated in water repellency, water retention, insulation in cold environments, reflectance of solar radiation in hot environments, herbivore deterrence, and many other functions (Brewer *et al.* 1991, Liakoura *et al.* 1997, Schreuder *et al.* 2001, Bickford 2016, Karabourniotis *et al.* 2020). Traits comprising leaf ecomorphs in LPT and SGE species may be brought together through selective pressures mediated by leaf physiology, shared genetic architecture of syndrome traits, or some relative contribution of each.

Viburnum jucundum and *V. lautum* occupy subtly distinct environments in the central highlands of Chiapas, where their geographic ranges broadly overlap at elevations between 1500 and 2800 m a.s.l. in the region surrounding San Cristóbal de las Casas (Donoghue 1982, Donoghue *et al.* 2022). *Viburnum lautum* (SGE) plants are shrubs (often 1 m tall or less in height) that grow in more open (lower canopy cover) pine-oak forests with higher daily temperature fluctuations than experienced by *V. jucundum* (LPT) plants, which are generally small trees (often 5 m or more in height) that tend to grow instead in more closed (higher canopy cover) cloud forests. During the drier winter months, *V. lautum* populations generally receive less cloud cover and moisture than *V. jucundum* populations, and they experience greater diurnal temperature fluctuations, including freezing temperatures on occasion (Donoghue *et al.* 2022). Differences in flowering time are known to maintain reproductive isolation between closely related species elsewhere in *Viburnum* (Donoghue 1980, 1982, Spriggs *et al.* 2019), but plants of these two species can flower at times throughout the wet season from April through September. However, their environmental niche differences are normally sufficient to spatially separate *V. lautum* and *V. jucundum* populations from one another and thereby prevent interbreeding.

The central plateau of Chiapas has been occupied by the Maya peoples for more than 2000 years, and in many areas the forests have long been modified through a variety of agricultural practices (Vogt 1969, Collier 1975). In some places patches of land have been maintained in a disturbed state for grazing, and it is in such disturbed ‘hybrid’ habitats (Anderson 1948) where the isolation of *V. jucundum* and *V. lautum* has sometimes broken down completely, and extensive hybrid swarms are encountered. One of these swarms was studied in detail by Donoghue (1982). Based on measurements of leaf and floral characters, he demonstrated the morphological intermediacy of many plants in this large population. Donoghue (1982) also carried out a series of controlled crosses, showing that these plants can interbreed and can be successfully backcrossed to parental phenotypes. Plants identified in the field as probable hybrids generally had lower pollen viability (12.9–88.8%) than did plants with ‘pure’ parental phenotypes (> 90% in both species), but this reduced viability did not appear to prohibit their reproduction (Donoghue 1982). And, although these plants were shown to set fewer fruit when selfed, some fruits were nevertheless produced by selfed

individuals, indicating that F_2 plants could also be present in these hybrid swarms.

Here, we confirm that hybrid swarms in several areas on the central plateau of Chiapas contain many individuals that are phenotypically variable and genetically admixed between *V. jucundum* and *V. lautum*. We also take an admixture mapping approach to associate leaf traits with genetic marker loci in the hybrids, uncovering the genetic architecture of individual leaf traits that characterize the parental leaf ecomorphs. Understanding the genetic architecture of leaf traits in these species is of particular interest because *V. lautum* and *V. jucundum* are not the only sister species pair in *Viburnum* that exhibit contrasting leaf syndromes. Within the *Oreiotinus* radiation, we have documented multiple instances of *Viburnum* sister species in which one species has LPT leaves and the other has SGE leaves (Donoghue *et al.* 2022). Insights from our work on this one species pair could further our understanding of the genetic basis of the parallel evolution of these contrasting leaf forms throughout the neotropical radiation of *Viburnum*.

MATERIAL AND METHODS

Sample collection

After extensive exploration and sampling over multiple field campaigns throughout central Chiapas, Mexico (Donoghue 1982, Donoghue *et al.* 2022), we located multiple populations of *V. jucundum* and of *V. lautum*, as well as mixed populations of *V. jucundum*, *V. lautum*, and individuals that, based on morphological assessments in the field, were putative hybrids (Fig. 1). In order to assess the morphological and genetic diversity across these species and their hybrids, we collected leaves from multiple individuals across six *Viburnum* populations in the highlands of central Chiapas: two populations of *V. jucundum* and two of *V. lautum* from sites where we expected gene flow between parent species to be limited or non-existent, and two populations containing putative hybrids (Fig. 1A, pins). Eight *V. jucundum* individuals were collected from Cerro Huitepec on the western edge of San Cristóbal de las Casas, along with one *V. jucundum* individual from Yalentay (~17 km west of San Cristóbal), for nine parental *V. jucundum* samples total. We also collected eight *V. lautum* individuals: four from Teopisca, ~33 km southeast of San Cristóbal and four from Reserva Ecológica y Orquideario Moxviquil on the northern edge of San Cristóbal. From two hybrid zones, located near Zinacantan (~11 km west of San Cristóbal) and Yashtinin (~28 km east of San Cristóbal), we collected 74 and 45 individuals, respectively. Given the relatively low level of leaf morphological variation within *V. jucundum* and *V. lautum* populations it is unlikely that these samples would be informative in identifying genetic variants associated with leaf form diversity. Therefore, we chose to sample more extensively within the hybrid swarms than across presumably non-admixed populations of *V. jucundum* and *V. lautum*. In addition to *V. jucundum*, *V. lautum*, and hybrid zone collections, we sampled three *Viburnum hartwegii* individuals for genetic sequencing to root our phylogenetic analysis. *Viburnum hartwegii* is a closely related species with IGE leaves that occupies generally lower elevations around the central plateau of Chiapas (Donoghue *et al.* 2022).

For leaf morphometric analyses, we pressed and dried at least five mature, fully-expanded leaves with minimal herbivore damage from each individual. These leaves were collected across multiple nodes along the branch and often across multiple branches per individual to provide an average value for each individual across the characters detailed below. Leaves were collected from non-reproductive branches to ensure that they did not develop within buds containing flowers, which could affect their morphology. In addition to leaves collected for morphological analyses, leaves from the same individuals were collected in silica gel for DNA extraction, and one branch was pressed as a voucher specimen, housed at the Peabody Museum Herbarium (YU) (for collection information see in [Supporting information, Table S1](#)).

Leaf phenotyping

Leaf trait measurements focused on characters that varied between previously documented *Viburnum* leaf ecomorphs ([Donoghue et al. 2022](#)) including leaf blade area, trichome density, and teeth along the leaf margin. We also included trichome branching characters and leaf length/leaf width measurements as these traits can also help to distinguish *V. jucundum* and *V. lautum* leaves in the field. Dried leaves used for morphological analysis were detached from branches and photographed against a standard herbarium sheet using a Nikon D810 DSLR camera fitted with a Nikkor 60 mm 2.8 G lens. Leaf blade area, length, and width were then measured using the ImageJ plugin LeafJ ([Maloof et al. 2013](#), [Rueden et al. 2017](#)). For trichome density measurements, we took a subset of three leaves from each individual and painted a 5 mm diameter circle on the adaxial and abaxial lamina at a point halfway up the midvein and halfway between the midvein and the margin, taking care to avoid major veins. We photographed this circle using a Zeiss Lumar V12 Stereoscope with Zeiss Axiocam 305 colour camera (Oberkochen, Germany) and counted the number of trichomes within the circle.

Stellate trichomes are the dominant trichome type for *V. jucundum* while simple trichomes are sometimes found on *V. lautum* individuals at very low densities. Stellate trichomes are also sometimes associated with leaf functional differences including increased herbivory deterrence and increased foliar water uptake ([Kariyat et al. 2018](#), [Li et al. 2023](#), [Araújo et al. 2024](#)). To explore patterns of trichome branching in our samples, we also used the subset of leaves used for trichome density measurements to classify trichome branching for 110 of our 119 hybrid swarm individuals. We inspected the entire surface of each leaf under a stereomicroscope (Motic SMZ-171) and scored trichomes as stellate, simple (with a single arm/cell), bifid (with two arms), or absent (glabrous) over the abaxial lamina, adaxial lamina, margin, and veins. Trichome branching (when trichomes were present) was then scored as a binary trait—stellate or not stellate.

To measure toothiness, we counted the number of teeth along the entire leaf margin of the same leaves used for trichome density measurements. In *Viburnum*, marginal teeth form very early during leaf development, before the leaf fully expands, determining blade area. Because of this, we chose to quantify toothiness as the total number of leaf teeth rather than using

tooth density (tooth number/leaf perimeter), which is not independent from blade area and determined later in development. All leaf trait measurements are available in [Supporting information, Table S1](#).

DNA extraction, RAD sequencing, and sequence assembly

In order to census the genetic variation across our samples and assess whether individuals growing in the hybrid swarm were indeed genetically admixed we sequenced a subset of each individual's genome using restriction site-associated DNA sequencing (RADseq). To do this, we extracted DNA from silica dried leaves using a modified CTAB protocol ([Healey et al. 2014](#)). Floragenex (Portland, Oregon, USA) digested the resulting DNA with *Pst*I to create 100-bp single-end libraries for RADseq ([Baird et al. 2008](#)) and sequenced each library on an Illumina Hi-seq 4000 at the University of Oregon (Eugene, Oregon, USA). Samples with more than 1 million reads were used in analyses going forward, yielding nine *V. jucundum*, eight *V. lautum*, and 119 hybrid swarm individuals which were also used for leaf phenotyping. We sequenced three *V. hartwegii* individuals using the same parameters to serve as an outgroup in later analyses.

RADseq reads were mapped to a *V. lautum* reference genome ([Donoghue et al. 2022](#)) using ipyrad ([Eaton and Overcast 2020](#)). The draft genome consists of 79 251 scaffolds totalling 3088.88 Mb in length. Individual scaffold length ranged from 1 to 2452.32 kb in length with 90% of the assembly length made up of scaffolds at least 47 kb long and 50% made up of scaffolds at least 250 kb long. This resulted in two reference-guided assemblies: one from parent individuals (individuals expected to be *V. jucundum* or *V. lautum* growing outside of the hybrid swarms) plus three *V. hartwegii* samples, and another including only hybrid swarm individuals. We also used ipyrad to assemble RADseq reads *de novo*, without matching them to the reference genome; one assembly for just parent individuals, one assembly for parent individuals plus three *V. hartwegii* samples, and another for just hybrid swarm individuals. Each individual-level analysis required a read depth of at least six reads per locus to be included and each variable site had to be present in at least four individual-level samples to be included in the global analysis (for ipyrad parameters and assembly information see https://github.com/mmoeglein/viburnum_admixture). This resulted in two different assemblies where RADseq data were aligned to the genome and three different assemblies where RADseq data were assembled *de novo*.

Phylogenetic analyses

To confirm the relatedness of our parental samples and ensure our correct identification of *V. lautum* and *V. jucundum* populations based on morphology, we used maximum likelihood to infer phylogenetic trees including our *V. lautum* and *V. jucundum* individuals as ingroup accessions with *V. hartwegii* samples included for rooting. Starting with the output from our reference-guided *V. lautum*, *V. jucundum*, and *V. hartwegii* assembly, we filtered all our assembled loci to retain those with a minimum coverage of at least 25% of all samples and removed any samples that contained data for less than 10% of all sites. Our filtered dataset was concatenated using the window_extractor

module in the ipyrad-analysis toolkit to produce a final alignment of 12 045 893 bp containing 136 677 single nucleotide polymorphisms (SNPs) with 60.03% missing data. We used RAxML v.8.2.12 (Stamatakis 2014) to perform the main analysis using the GTRCAT evolutionary model, running 100 repetitions for bootstrap support. We also ran a cross-validation reconstruction using IQtree v.2.2.0.3 (Minh *et al.* 2020) using the model K3Pu+F+I+G4 suggested by ModelFinder (Kalyaanamoorthy *et al.* 2017), running 1000 replicates to get ultrafast bootstrap support values [UFBoot (Hoang *et al.* 2018)] and 1000 replicates to get the SH-like approximate likelihood ratio test support [SH-aLRT (Anisimova *et al.* 2011)]. Reproducible material for phylogenetic analyses can be obtained at <https://github.com/camayal/admixture-viburnum>.

Genome-wide ancestry

Genome-wide ancestry was estimated using STRUCTURE, which assigns individuals to populations based on their inferred global ancestry while allowing for admixture (Pritchard *et al.* 2000, Porras-Hurtado *et al.* 2013, Pereira *et al.* 2020, Larson *et al.* 2021). Output files containing variable sites from *de novo* RAD-seq assemblies for just the parent species samples, parent plus *V. hartwegii* samples, and hybrid swarm samples were run through STRUCTURE assuming two or three assigned parental populations ($K = 2$ or 3). For each STRUCTURE run, we had a burn-in of 50 000 runs followed by 100 000 replicates. This process was repeated three times providing three replicates for each value of k that were then combined using CLUMPP (Jakobsson and Rosenberg 2007) to derive the final ancestry estimates. We plotted these results in toyplot (<https://github.com/sandialabs/toyplot>) with the proportion of a given colour in each bar corresponding to the inferred proportion of ancestry in an individual.

Admixture mapping

Admixture mapping is a form of association mapping that allows us to test for relationships between DNA sequence variation and trait variation in our samples. To perform association mapping, we started with the variant call format (vcf) output files from the hybrid-swarm-only reference-mapped RAD-seq assembly from ipyrad (Eaton and Overcast 2020) described above. We filtered this output, requiring any variable site to be present in at least 80% of the samples, using VCFtools (Danecek *et al.* 2011) and imputed missing variants using BEAGLE (Browning *et al.* 2018). We used PLINK (Purcell *et al.* 2007) to convert the imputed vcf file from BEAGLE to binary pedigree (PED) format for association mapping. Trait measurements and imputed markers were then used as input for association mapping in GEMMA (Zhou and Stephens 2012). For continuous traits including blade area, blade length/width, abaxial trichome density, and adaxial trichome density, we ran GEMMA using the univariate linear mixed model with significance computed as a likelihood ratio test. For binary traits including the presence or absence of stellate trichomes on the abaxial lamina, adaxial lamina, leaf margins, and leaf veins, we ran GEMMA using the linear model with significance computed as a likelihood ratio test. GEMMA results were plotted using the R package qqman (Turner 2018), using the qqman default thresholds for genome-wide significance [$-\log_{10}(5 \times 10^{-8})$] and suggestive significance [$-\log_{10}(1 \times 10^{-5})$].

Relationships between traits and individual alleles were plotted using ggplot2 (Wickham 2016). Details of our usage of each of the packages noted above are provided at https://github.com/mmoeglein/viburnum_admixture, a repository for extended methods.

Gene annotation

Due to its large size and fragmented structure, annotation of the *V. lautum* genome is currently incomplete. Therefore, in order to identify genes potentially implicated in leaf differences, we structurally and functionally annotated windows of 20 kb before and 20 kb after every significant SNP uncovered by the GEMMA analysis. We carried out the structural annotation using MAKER v.3.01.03 (Cantarel *et al.* 2008). For this, we ran two rounds of annotation. The first round included as primary evidence the assembled transcriptome of *V. lautum* and a database of repeats found in the draft genome. We used Trinity v.2.9.1 (Grabherr *et al.* 2011) to assemble RNA sequences from root, leaf, and meristem tissue. We used RepeatMasker v.4.1.0 (Smit *et al.* 2021) for the entire genome to build a repeat database. In the second round, we enhanced our structural annotation by adding more evidence from *Arabidopsis thaliana* {protein sequences [Araport11 protein list, (Berardini *et al.* 2015)]} and the more closely related *Lonicera japonica* [structural annotation and transcriptome (Pu *et al.* 2020)]. Configuration files for each round as well the general feature format (GFF) files are available in our repository for reproducibility (https://github.com/camayal/admixture-viburnum/tree/main/Analyses/Gene-annotation/Structural_annotation/Results/MAKER/20kwindows). We carried out the functional annotation using Blast2Go v.6.0.3 (Conesa *et al.* 2005) to associate gene ontology (GO) terms for the predicted proteins found in the structural annotation. We used the Swiss-Prot (UniProt Consortium 2019) and InterPro (Paysan-Lafosse *et al.* 2023) collection of databases in Blast2Go to obtain the GO terms and merge them into a single matrix. Finally, we evaluated if the predicted and annotated gene models contained any of the significant SNPs identified by the GEMMA analysis. We then did a literature search for the protein descriptions and names to associate reported gene functions with our studied traits. All the scripts for window extraction and SNP evaluation can be found in our repository for reproducibility <https://github.com/camayal/admixture-viburnum>.

RESULTS

Phylogenetic relationships

We found strong support for reciprocal monophyly between *V. lautum* and *V. jucundum* (Fig. 1B). *Viburnum lautum* contained two clades corresponding to the two collection localities, Teopisca and Moxviquil. The *V. lautum* reference genome was generated from a *V. lautum* individual collected from the Teopisca population and, as expected, it clustered with the rest of the Teopisca sequences (Fig. 1B). The *V. jucundum* sample from Yalentay (MKM_11) was nested within the *V. jucundum* samples from Huitepec. Bootstrap support values at all nodes were 99 or 100 with the exception of a few nodes within the *V. jucundum* clade, which had a support value of 0. Overall, our phylogenetic analyses supports the relationship between *V. lautum* and

V. jucundum as distinct sister species consistent with previous studies (Landis et al. 2021, Donoghue et al. 2022).

Divergence in leaf phenotypes

Viburnum lautum and *V. jucundum* leaf traits contrasted strongly across most characters examined. *Viburnum lautum* had significantly smaller leaves with fewer marginal teeth and sparser trichome densities on both sides of the leaf than *V. jucundum* (Fig. 2A, C, D, E left panels; Supporting information, Fig. S1A, C, D, E; Welch's *t*-test $P < .001$). Though *V. lautum* did have more narrow, elongate leaves than *V. jucundum* on average, these differences were not statistically significant (Fig. 2B; Supporting information, Fig. S1B). Only one *V. lautum* individual had stellate trichomes and then only on its adaxial lamina, while all *V. jucundum* samples had stellate trichomes on both sides of the leaf (Supporting information, Fig. S4).

Leaf trait values for some hybrid swarm individuals closely matched our *V. lautum* and *V. jucundum* parental samples (Fig. 2A–E, grey squares within yellow and pink bands). Many others, however, exhibited intermediate or sometimes extreme values for all traits relative to the parent species samples (Fig. 2A–E, grey squares falling outside of yellow and pink bands). The highest recorded abaxial trichome densities, adaxial trichome densities, and number of marginal teeth in this study all belonged to individuals from the hybrid swarms (Fig. 2C–E; Supporting information, Fig. S1). As many *V. lautum* individuals completely lacked trichomes or leaf teeth, lower extremes in these traits were not possible in hybrid swarm individuals. Hybrid swarm traits on average were intermediate between, though also significantly different from *V. lautum* and *V. jucundum* leaf traits with the exceptions of adaxial trichome density, which was not significantly different from *V. jucundum* trait values, and blade length/width ratio, which was not significantly different across hybrid swarm individuals or parent species (Supporting information, Fig. S1).

Of the 110 hybrid swarm individuals assessed for trichome branching, some individuals did not have any stellate trichomes while others had stellate trichomes across both sides of the leaf, and on the margins and veins (Supporting information, Fig. S5). We also found variation between these extremes, such as individuals with stellate trichomes everywhere except the leaf veins or individuals with stellate trichomes only on the abaxial lamina (Supporting information, Fig. S5). Hybrid swarm individuals with stellate trichomes on the abaxial or adaxial leaf surface also had higher trichome densities on the abaxial or adaxial leaf surfaces (Supporting information, Fig. S3).

Ancestry and admixture in *Viburnum* populations

The parental species *V. lautum* and *V. jucundum* had largely separate ancestry based on our STRUCTURE analysis with $K = 2$ ancestral populations (Fig. 2F), although there was a small proportion of *V. jucundum*-associated ancestry found in the *V. lautum* samples from Teopisca (4–12% per individual), possibly reflecting a small amount of introgression or simply population differentiation between the Teopisca and Moxviquil specimens. When we analysed these samples assuming three ancestral populations ($K = 3$, Supporting information, Fig. S7B) we found clearer differentiation between *V. lautum* from Teopisca and Moxviquil, but also some potential shared ancestry with a

few *V. jucundum* samples (Supporting information, Fig. S7B). When we added *V. hartwegii* to our *V. lautum* and *V. jucundum* STRUCTURE analyses, we found differentiation between *V. hartwegii* and *V. jucundum* demonstrating separate ancestry at $K = 2$ while *V. lautum* appeared intermediate (Supporting information, Fig. S6A). At a value of $K = 3$ all three species were clearly differentiable (Supporting information, Fig. S6B) with *V. lautum* and *V. jucundum* ancestry in similar proportions to the $K = 2$ STRUCTURE run that did not include *V. hartwegii* (Fig. 2F). Overall, these results suggest little if any admixture between isolated populations of these species outside of hybrid swarm environments and affirm the ability of STRUCTURE to distinguish between these species.

When estimating genome-wide ancestry of the hybrid swarm individuals, results from a STRUCTURE analysis for $K = 2$ ancestral populations (Fig. 2F) suggest a gradient of admixture across hybrid swarm samples. Some individuals collected from the hybrid swarms are inferred to have pure parental genomes, suggesting that some pure *V. lautum* (Fig. 2F) and *V. jucundum* (Fig. 2F) individuals are present within the otherwise well-admixed hybrid populations. Across hybrid swarm samples, *V. lautum* ancestry is slightly more prevalent than *V. jucundum* ancestry (Fig. 2F). A STRUCTURE analysis of the hybrid swarm populations given $K = 3$ uncovered a proportion of *V. lautum* ancestry associated with the Yashtinin population (Supporting information, Fig. S7B) but otherwise provided similar estimates of *V. jucundum* ancestry as the $K = 2$ analysis.

Association mapping for hybrid leaf traits and corresponding phenotypic changes

GEMMA analyses identified significant relationships between genetic markers and all traits examined except for blade length/width (Fig. 3A), a trait that did not vary significantly between *V. jucundum*, *V. lautum*, and hybrid swarm populations (Supporting information, Fig. S1B), and was therefore unlikely to allow for the detection of underlying genetic associations. There were 37 markers with genome-wide significance for blade area (Fig. 3A), two for marginal teeth (Fig. 3A), four for abaxial trichome density (Fig. 3B), and one for adaxial trichome density (Fig. 3B), along with many more markers with suggestive significance for each of these traits (Fig. 3). There were also markers of genome-wide significance associated with stellate trichomes: two for adaxial trichomes (Fig. 3C) and one each for abaxial (Fig. 3C) and vein (Fig. 3C) trichomes; margin trichomes only had markers with suggestive significance (Fig. 3C).

Three markers with genome-wide or suggestive significance for marginal teeth also had genome-wide or suggestive significance for blade area (Fig. 3A, black ovals). The one marker most strongly associated with adaxial trichome density also had a significant association with abaxial trichome density (Fig. 3B, black oval). Markers associated with the presence of stellate trichomes were also shared across different regions of the leaf (Fig. 3C, black ovals). Though there were shared significant markers between some traits (blade area and marginal teeth, abaxial and adaxial trichome density, stellate trichomes across leaf regions), no markers strongly or suggestively associated with a pair of traits were shared across all traits (i.e. no significant markers are shared across traits depicted in Fig.

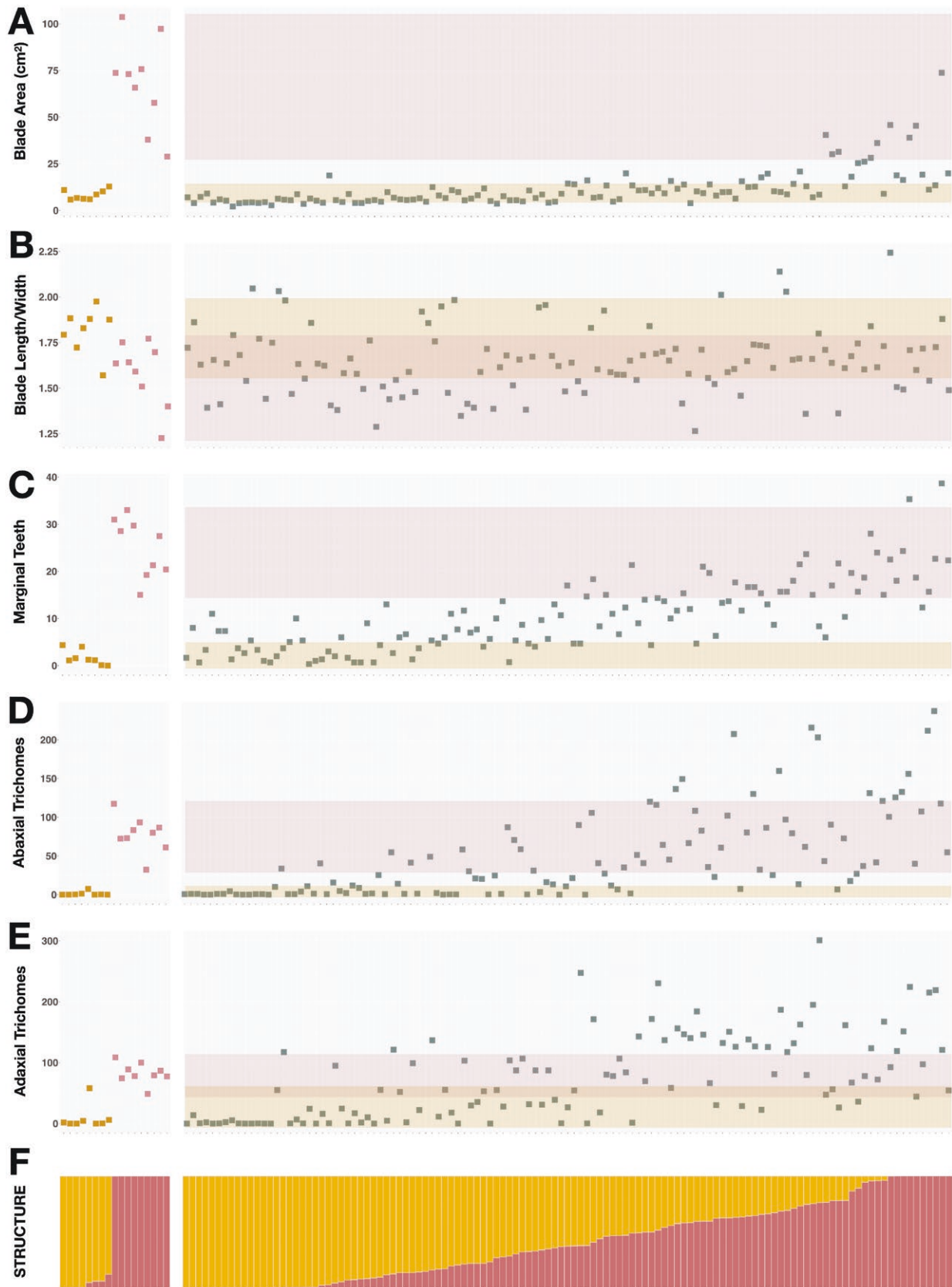


Figure 2. Average trait values for individual plants (A–E) with *V. lautum* (yellow) and *V. jucundum* (pink) displayed on left and hybrid swarm individuals plotted on the right in grey. Yellow and pink boxes overlapping hybrid swarm plots show the range of trait values for *V. lautum* and *V. jucundum*, respectively, in relation to hybrid swarm trait values. Genome-wide ancestry (F) as calculated by STRUCTURE at $k = 2$ with *V. lautum* and *V. jucundum* individuals on the left represented by predominantly yellow or pink bars, respectively, and all hybrid swarm individuals on the right showing varying amounts of *V. lautum* (yellow) or *V. jucundum* (pink) ancestry.

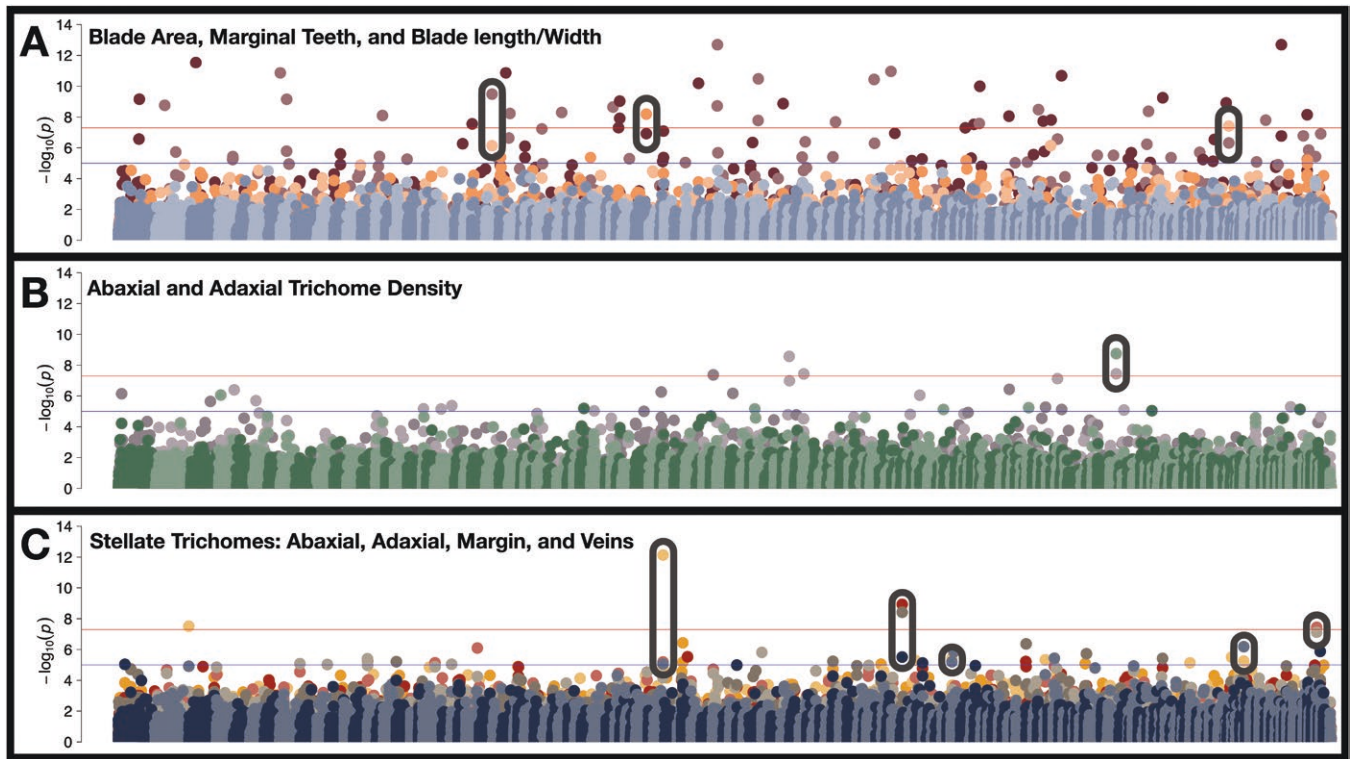


Figure 3. Manhattan plots with each point representing a SNP between the hybrid swarm sequences and the reference genome sequence. Shading of points alternates by genome scaffold with scaffolds ordered from largest to smallest. Points with a greater $-\log_{10}(p)$ value have a stronger correlation with a given trait: points located above the blue line have suggestive significance while points above the red line have genome-wide significance. Three plots overlaid in panel (A) represent genome-wide associations for blade area (red points), number of marginal teeth (orange points), and blade length/blade width (blue points). Three SNPs associated with blade area are also associated with the number of marginal teeth (grey ovals). Two plots overlaid in panel (B) represent genome-wide associations for abaxial trichome density (grey points) and adaxial trichome density (green points). One SNP associated with adaxial trichome density is also associated with abaxial trichome density (grey oval). Four plots overlaid in panel (C) represent genome-wide associations with the presence of stellate trichomes on four different leaf surfaces: abaxial lamina (yellow points), adaxial lamina (red points), abaxial leaf veins (grey points), and leaf margins (blue points). Five SNPs were associated with at least two of these traits (grey ovals). There was no overlap in significant SNPs between the traits represented in panels (A), (B), and (C).

3A–C). Given the reduced representation inherent in RADseq and the fragmented state of the *V. lautum* draft genome we are likely underestimating the total number of significant markers for each trait, and we address the limitations of this approach further in the Discussion.

As expected, when we compared trait values for individuals carrying the *V. lautum* reference allele to those with the alternative allele for the markers most significantly associated with the focal traits, we found that the individuals carrying the alternative allele had more *V. jucundum*-like phenotypes, with larger blade areas, more marginal teeth, and higher trichome densities (Fig. 4). The only exception was locus 88 589, position 166, a marker significantly associated with marginal teeth, where the reference allele was actually associated with increased toothiness (Fig. 4B, third boxplot from left). We analysed phenotypic changes for the top five most significant SNPs across tooth number, abaxial trichome density, and adaxial trichome density, and we analysed phenotypic change for the top seven SNPs associated with increases in blade area as three of the top seven blade area SNPs only occurred in the same two individuals (Fig. 4A, first boxplot from left). The top five significant SNPs for various traits were common and often shared across different subsets of hybrid

swarm individuals, with only four individuals lacking any of the alternative alleles (Supporting information, Fig. S8).

Annotated genes associated with significant markers

After annotating 20 kb windows around each marker locus with a significant trait association from our admixture mapping analysis (59 significant markers in total), we recovered 177 functionally-annotated genes with 3393 GO terms across 55 windows (Supporting information, Table S2). For 27 of the total 95 gene models recovered, the SNPs from our RADseq assembly fell within the gene itself (Supporting information, Table S3). After inspecting putative gene function data and conducting a broader literature search, we found some promising candidate genes with evidence for leaf developmental and morphological functions (discussed further below).

DISCUSSION

Parent species ecomorphs, hybrids, and phenotypic accessibility

Our morphological characterization of the sister species *V. lautum* and *V. jucundum* reiterates the findings from Donoghue

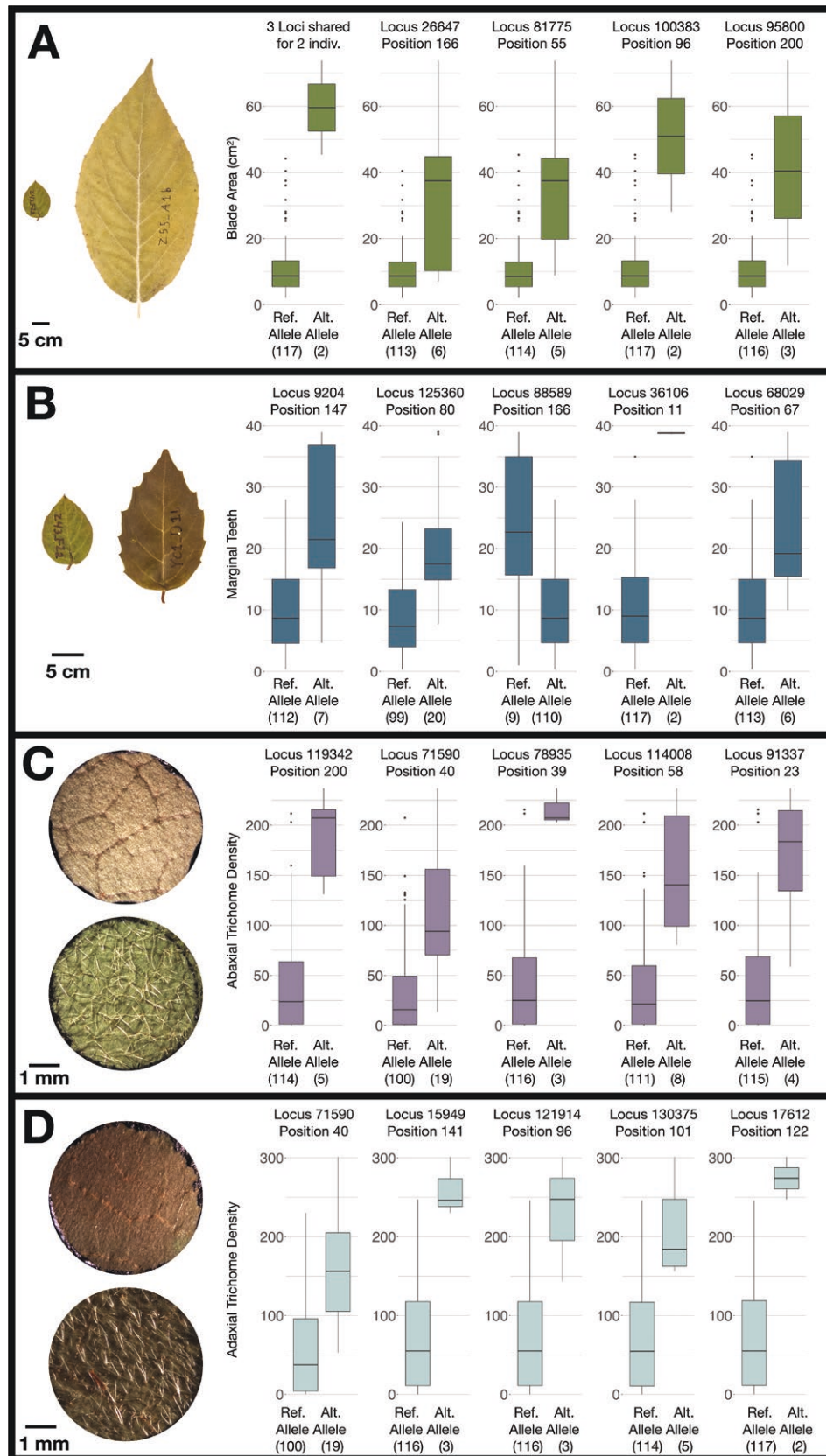


Figure 4. Boxplots depicting trait values for hybrid swarm individuals carrying the reference allele vs. the alternative allele for the top five loci of largest effect on blade area (A), number of marginal teeth (B), abaxial trichome density (C), and adaxial trichome density (D). Numbers under boxes indicate how many individuals out of the 119 total hybrid swarm individuals carried the reference genome allele (box on left) or the alternative allele (box on right). Images to the left of plots depict variation between hybrid swarm individuals in blade area (A), number of marginal teeth (B), abaxial trichome density (C), and adaxial trichome density (D). Images on left are from hybrid swarm individuals demonstrating phenotypic extremes in the respective traits.

et al. (2022): *V. lautum* leaves are relatively small, glabrous, and entire-margined, aligning with the small, glabrous, entire (SGE) ecomorph, whereas *V. jucundum* leaves are relatively large and stellate pubescent with toothy-margins, corresponding to the large, pubescent, toothed (LPT) ecomorph. We also expand on the previous morphological descriptions in Donoghue *et al.* (2022), quantifying trichome density and evaluating trichome branching across the entire leaf surface. In *V. lautum*, we counted very low trichome densities on the abaxial and adaxial lamina (Fig. 2D, E) and these trichomes were very rarely stellate (Supporting information, Fig. S4). *Viburnum lautum* leaves were often entire-margined but some individuals had a few marginal teeth (Fig. 2C). In spite of this variation, there was still clear differentiation from *V. jucundum* across all traits that delineate the ecomorphs (Supporting information, Fig. S1). This provides further evidence of the clear divergence of ecomorphs within regions of endemism across the replicated radiation identified by Donoghue *et al.* (2022) and a more nuanced description of trichome character divergence in particular.

While our parent species, *V. lautum* and *V. jucundum*, exhibited clear morphological differentiation, the hybrid swarm individuals demonstrated a wide phenotypic range and a reshuffling of leaf characters. Within the hybrid swarm, we found some individuals that matched our *V. lautum* and *V. jucundum* parent samples, both in phenotype and in ancestry inferred from our STRUCTURE analysis (Fig. 2). Other hybrid swarm individuals, with intermediate leaf phenotypes, were admixed with varying levels of genetic intermediacy (Fig. 2). We also found multiple instances of transgressive phenotypes in the hybrid swarms including very high trichome densities, small blade areas, and a greater incidence of marginal teeth in some individuals (Fig. 2; Supporting information, Fig. S2). Our documentation of transgressive phenotypes in hybrids is in line with previous work (Rieseberg *et al.* 1999, Stelkens and Seehausen 2009, Stelkens *et al.* 2009, Yakimowski and Rieseberg 2014), though after extensive exploration and identification of *Viburnum* populations throughout Chiapas (Fig. 1A; Donoghue *et al.* 2022) these alternative phenotypes do not appear to colonize new environments beyond the hybrid swarm or form new species as has been found in other systems (Rieseberg *et al.* 2007, Feller *et al.* 2020).

The Zinacantan hybrid swarm population was first observed over 40 years ago (Donoghue 1982) and the continuous gradation of inferred parental ancestry found in our STRUCTURE analysis suggests that the hybrid swarm has persisted for many generations. Hybrid environments can sometimes select for traits in hybrid offspring outside of those favoured in parental populations and their environments (Anderson 1948, Estévez Manso Galán *et al.* 2024) and it is possible that some traits or trait combinations present in these hybrids may be particularly beneficial in the high-light, high-disturbance environment of the swarm, promoting its continued existence. Small leaves are more common in hotter, drier environments (Wright *et al.* 2017) and higher trichome densities can also increase solar reflectance (Liakoura *et al.* 1997, Karabourniotis *et al.* 2020). It would be interesting to test, perhaps through a reciprocal transplant experiment, if the combination of small, hairy leaves is particularly beneficial in the swarm environment though they are uncommon outside of it.

This work focuses on leaf traits, but we know from previous studies that reproductive traits vary across the hybrid swarm individuals as well (Donoghue 1982) and this variation may perpetuate the generation of hybrids. Our observations in the field suggest that *V. lautum* and *V. jucundum* may have more irregular, extended, and overlapping flowering times in disturbed higher-light environments, which describe our current hybrid swarm localities. We also suspect that parents and hybrids may commence flowering at earlier ages in such environments. Such shifts in flowering behaviour can, in turn, promote the continued generation of hybrid plants (Tochigi *et al.* 2021) with trait combinations that may not be as successful outside of these heavily disturbed settings.

Even for hybrid swarm individuals exhibiting leaf trait values within the parental range, we uncovered multiple instances where the traits that differentiate the neotropical *Viburnum* ecomorphs were dissociated and reshuffled in the hybrids. For example, multiple hybrid swarm individuals have very small leaves with toothy margins while others have many marginal teeth but are relatively glabrous (Figs 1C, 2; Supporting information, Fig. S2). Although parallel evolution of a set of morphological traits alone is not proof of an adaptive syndrome (Sinnott-Armstrong *et al.* 2022), the potential trait dissociation documented here suggests that the LPT and SGE ecomorphs may be favoured by selection in particular environments, and that other accessible permutations of these traits are selected against and do not persist outside of hybrid swarms. The phenotypic variation generated by the dissociation of individual leaf traits can also help us to identify the genetic architecture that may have influenced ecomorph assembly in the first place.

Genetic architecture of leaf traits

In view of the small scaffold sizes in our genome assembly, we caution against over-interpreting our results with respect to the positions of marker loci in the genome. In particular, some significant markers for blade size may be closer together than they appear, and the same could be true for margin and trichome characters as well. It is also possible that the genomic subsampling inherent in RADseq underestimated the number and location of genomic regions associated with our traits of interest and therefore also possible that some of these missed regions could be shared across traits. The use of RADseq for association mapping only directly sequences a small subset of the genome and relies on linkage between sequenced loci and relevant associated genomic regions, therefore this approach may miss significant regions located farther from available markers (Lowry *et al.* 2017). However, regions of the genome with linkage to markers of interest allows for a wider census area than just the markers themselves and the use of RADseq in association mapping has uncovered associated genes in a variety of organisms (Brelsford *et al.* 2017, McKinney *et al.* 2017, Yamashita *et al.* 2020). We do not have an accurate estimate of linkage disequilibrium in our hybrid swarm and currently do not know if we may be missing trait-associated loci, including loci that may indicate pleiotropy between traits.

Though there are limitations to association testing with RADseq data, this approach still uncovered individual marker loci of significant effect for most of our traits (Figs 3–4). Blade

area was significantly associated with 37 markers while many more were suggestive and may play a minor role in specifying blade size, consistent with many loci of small additive effect and potentially providing multiple paths to blade size variation. Admixture mapping studies investigating leaf traits in *Populus* also found multiple marker loci associated with blade area interspersed across the genome (Lindtke *et al.* 2013). We note, however, that this work involved considerably fewer markers than examined here and a subsequent study of *Populus* failed to identify markers with a significant blade area association (Bresadola *et al.* 2019). Leaf size can also be a particularly plastic trait (Mähler *et al.* 2020, Stotz *et al.* 2022), potentially making it more challenging to compare between studies and different environmental conditions.

Only two loci were significantly associated with marginal teeth, though four others had a suggestive correlation. This relatively simple genetic architecture with few associated loci could reflect missing associations due to our genomic subsampling, but even in organisms with more complete genomic resources it is common to find traits with relatively few associated regions (Boyko *et al.* 2010, Hager *et al.* 2022). Interestingly, the two significant loci for marginal teeth also had a suggestive correlation with blade area. This could mean that although blade area and marginal teeth are not always associated (i.e. blade area may increase while maintaining an entire margin), there are some paths to increased blade area that are more tightly correlated with the development of marginal teeth.

Trichome-density-associated loci comprised a completely different set of markers than those associated with blade and margin characters, though there was overlap for significant markers associated with abaxial and adaxial trichome density. This suggests that increases in trichome density on one side of the leaf can be linked to an increase in trichome density on the other side as well. Significant loci associated with trichome branching also comprised a completely different set of markers that did not overlap with the trichome density markers or the size and teeth markers. The lack of overlap in significant markers associated with stellate trichomes may indicate genetic independence between trichome density and trichome morphology.

Overall, it appears that while blade area and leaf teeth, trichome densities, and trichome branching can and do vary independently, there are some genomic associations that may render some combinations more likely. Blade area and marginal teeth appear mostly independent, but some alleles promoting toothiness also are associated with larger leaf blades. However, our mapping analysis implies that there are many ways to vary blade area without influencing marginal teeth. Trichome densities may be even more free to vary independently of blade size and marginal teeth, given the apparent lack of shared loci. Abaxial and adaxial trichome densities do share a significant marker and are very often phenotypically correlated (Supporting information, Fig. S2F).

While our admixture mapping approach may not uncover all genomic regions relevant to our traits of interest, the positive evidence of overlapping genetic architecture such as that between blade area and leaf teeth could promote non-independence between these traits. Other traits that appear to have wholly separate genetic architecture may have overlapping loci that we did

not uncover in our admixture mapping analysis. Nevertheless, this interplay between correlation and independence for different sets of traits echoes earlier trait mapping work in *Mimulus cardinalis* populations with different life-history syndromes (Nelson *et al.* 2021), and in benthic and limnetic sticklebacks (Arnégard *et al.* 2014), where there are genetic associations within some trait modules and genetic independence between others.

Candidate genes associated with leaf traits

Individual marker loci associated with genetic differences between *V. jucundum* and *V. laetum* were sometimes found near, or even within, genes with known significance for leaf development in other plants, providing an avenue to further explore their potential functions. Among the many markers with significant association with blade area, one locus (scaffold 61434, SNP 8380992) was located within a gene showing strong similarity to *Arabidopsis thaliana* 7-dehydrocholesterol reductase, a gene previously shown to be associated with a more compact growth architecture and smaller leaves in cucumber (Zhang *et al.* 2021). Another marker significantly associated with blade area (scaffold 17365, SNP 7371028) was located near a gene showing strong sequence similarity with the *A. thaliana* Myosin-7 gene, which in turn codes for a protein associated with the governance of cytoplasmic streaming velocity, where increased streaming velocities during development are associated with larger leaf sizes (Tominaga *et al.* 2013). A marker showing significant association with both blade area and marginal teeth (scaffold 63432, SNP 2670146) fell within a gene identified as histidine kinase 4, which also governs a variety of leaf morphological traits (including leaf size and shape in *A. thaliana*) by facilitating the migration of signalling molecules across tissues (Nishimura *et al.* 2004, Deng *et al.* 2010).

Regions of the *Viburnum* genome associated with trichome density and branching also contained genes previously implicated in leaf development. The 26S proteasome contributes to abaxial/adaxial patterning in *A. thaliana* and other plants (Huang *et al.* 2006) and one of our markers associated with abaxial trichome density (scaffold 23621, SNP 2087672) was located near a gene encoding a regulatory subunit of 26S (26S proteasome non-ATPase regulatory subunit 2, Supporting information, Table S2), suggesting a possible regulatory mechanism for specifying trichome patterning on either side of the leaf. A marker associated with increases in abaxial trichome density (scaffold 4057, SNP 1779621) was also located within a gene associated with morphological determination and signalling, MEKK3, which is involved in the phosphorylation of PIN1, an auxin transporter that governs leaf patterning (Jia *et al.* 2016). Earlier work on the genetic mechanisms underlying trichome development often focused on model systems where endoreduplication plays a central role (Fambrini and Pugliesi 2019). *Viburnum* trichomes are not endoreduplicated structures and thus many previously identified candidate genes may not be relevant here.

Though suggestive and in need of validation, the annotated genes corresponding to our significant markers provide some exciting avenues to further explore how leaf syndromes evolved repeatedly across the *Oreinotinus* clade. The candidate genes

identified here provide an excellent starting point as we uncover genetic changes in other *Viburnum* species pairs with contrasting leaf forms.

Leaf ecomorphs in *Viburnum*

In view of what we have discovered about the underlying genetic architecture of the traits that distinguish leaf ecomorphs in neotropical *Viburnum*, should we be more or less surprised by their repeated evolution? We do not yet know if our findings for the *V. jucundum*-*V. lautum* sister species pair can be generalized to other species with these contrasting leaf forms, but given their close relationships, similar genome sizes, and shared ploidy levels (Moeglein et al. 2020, Donoghue et al. 2022), we suspect that they will share elements of the underlying genetic architecture that we have described here. Given the relatively large number of markers associated with blade size, this trait may be easier to change additively and incrementally, with multiple alleles involved in potentially multiple pathways to a larger leaf. Alternatively, we cannot rule out that there could be a number of loci of larger effect and it may be relatively easy to acquire any of them. Either of these options could render different leaf size especially evolutionarily labile. In addition, the potential links between blade size and toothiness could make it more likely that larger leaves will bear marginal teeth.

It is possible that our use of RADseq to generate markers for admixture mapping did not uncover every genomic region underlying our traits of interest, including regions that may influence multiple traits. However, given the non-overlapping sets of discovered markers related to trichome density vs. trichome branching vs. blade and margin characters, large and toothy leaves that are also hairy seem less likely to arise simply as a function of pleiotropy. Yet it is very rare for small-leaved *Viburnum* species to be especially hairy (the Mexican *Viburnum loeseneri* in *Oreiotinus*, and the Asian *Viburnum utile* in the distantly related *Euviburnum* clade being two notable exceptions), and throughout the *Oreiotinus* radiation the species with the largest leaves always have leaves densely covered with hair (Donoghue et al. 2022). Based on our knowledge of genetic architecture alone, it should be just as likely to find large, toothed, pubescent leaves as it would be to find large, toothed, glabrous ones, but the latter are unknown. This suggests a role for selection, i.e. that LPT or SGE forms are advantageous in certain contexts. Consistent with this adaptation hypothesis, and as noted above, *V. lautum* and *V. jucundum* do occupy subtly but consistently different environmental niches on the central plateau of Chiapas (Donoghue et al. 2022). And, similar environmental differences appear to characterize some (though not all) of the sister species with these two leaf syndromes in other montane areas in the neotropics (Donoghue et al. 2022).

Many potential climatic and ecological factors have been identified as potential factors that select for different leaf sizes (Wright et al. 2017, Pereira and Des Marais 2020), leaf margins (Bailey and Sinnott 1916, Givnish 1987, Schmerler et al. 2012), and patterns of leaf pubescence (Schreuder et al. 2001, Agrawal et al. 2009, Bickford 2016), but to our knowledge there are no studies that have identified the genetic or environmental factors underlying specific combinations of these traits such as we see in the *Viburnum* leaves. We have speculated that SGE leaves may better accommodate overheating during the summer and

overchilling during the winter (Donoghue et al. 2022). And, the large leaves of the LPT species may aid in light capture in the forest understory, and their dense pubescence may aid in gas exchange by keeping water off of the leaf surface. Presence of leaf teeth are more difficult to explain in these plants, but we have hypothesized elsewhere that these may relate to the duration and extent of leaf development inside of resting buds (Edwards et al. 2016). Our finding here of a possible genetic connection between leaf area and toothiness opens the possibility that selection for increased size may have promoted tooth production. Clearly, more functional and developmental studies are needed to understand our ecomorphs as adaptive syndromes (Sinnott-Armstrong et al. 2022), but our preliminary knowledge of the genetic architecture underlying individual traits allows us to better hypothesize how syndromes may have been assembled.

CONCLUSION

Our analyses confirm extensive past and ongoing hybridization between *V. jucundum* and *V. lautum* in persistently disturbed areas. They also provide a clear link between genomic ancestry and phenotypic variation in leaf traits. Although we have shown that leaf traits have variable genomic underpinnings and are largely correlated with separate genomic regions, a few key genomic regions are shared between certain traits, suggesting that some traits are more likely to evolve in a correlated fashion, such as larger leaf size and toothed margins.

SUPPORTING INFORMATION

Supplementary data is available at *Evolutionary Journal of the Linnean Society* online.

Figure S1. Boxplots comparing leaf trait values for individuals from *V. lautum* populations, *V. jucundum* populations, and hybrid swarm populations.

Figure S2. Scatterplots showing relationships between trait values across *V. lautum* populations (Moxviquil and Teopisca), *V. jucundum* populations (Huitepec and Yalentay), and hybrid swarm populations (Yashtinin and Zinacantan).

Figure S3. Boxplots comparing trichome densities for hybrid swarm individuals with and without stellate trichomes.

Figure S4. Scatterplots showing relationships between trait values across *V. lautum* populations (yellow, Moxviquil and Teopisca), *V. jucundum* populations (pink, Huitepec and Yalentay), and hybrid swarm populations (grey, Yashtinin and Zinacantan) with shapes indicating the presence (triangles) or absence (circles) of stellate trichomes on the abaxial lamina or adaxial lamina.

Figure S5. Venn diagram showing number of hybrid swarm individuals with stellate trichomes across different regions of the leaf (veins, margins, abaxial lamina, and adaxial lamina).

Figure S6. STRUCTURE plots showing proportion of inferred ancestry assuming two ancestral populations ($K = 2$) or assuming three ancestral populations ($K = 3$) for samples from predominantly *V. lautum*, *V. jucundum*, and *V. hartwegii* populations.

Figure S7. STRUCTURE plots showing proportion of inferred ancestry assuming two ancestral populations ($K = 2$) or assuming three ancestral populations ($K = 3$) for samples from

predominantly *V. lautum* and *V. jucundum* populations and from the hybrid swarm.

Figure S8. Venn diagram showing number of hybrid swarm individuals with one or more of the top five most significant genetic variants associated with blade area, number of marginal teeth, abaxial trichome density, and/or adaxial trichome density.

Table S1. A list of all collection locality and specimen naming information for the *V. jucundum*, *V. lautum*, and hybrid swarm specimens used in this study.

Table S2. Functional annotation results for windows around loci with significant trait associations.

Table S3. Structural annotation results for windows around loci with significant trait associations.

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DATA AVAILABILITY

The *V. lautum* draft genome assembly and transcriptome sequences used for gene annotation are deposited in the NCBI sequence read archive (PRJNA830847). RAD sequencing data associated with this study is also available through the NCBI sequence read archive (PRJNA1081898). The GitHub repository <https://github.com/camayal/admixture-viburnum> contains analyses and data related to all phylogenetic and gene annotation analyses in this study while the further GitHub repository https://github.com/mmoelein/viburnum_admixture contains all scripts and leaf trait data related to RAD-sequence assembly, population structure analyses, genome-wide association, and leaf trait relationships.

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